

Can the Fed control inflation and maintain full employment?

Twenty-five years of inflation, interest rates and unemployment

Muhammad Suffyan Khan

DATA 608: Knowledge and Visual Analytics
CUNY School of Professional Studies · Fall 2026

25 years

September 2001 through August
2026

The measures and the question

02

300

monthly observations
September 2001–August 2026

3

series from BLS and FRED

1

question about the Fed's
dual mandate

Price stability

CPI index converted to 12-month inflation. The Fed targets 2%
PCE inflation, a different price measure.

Maximum employment

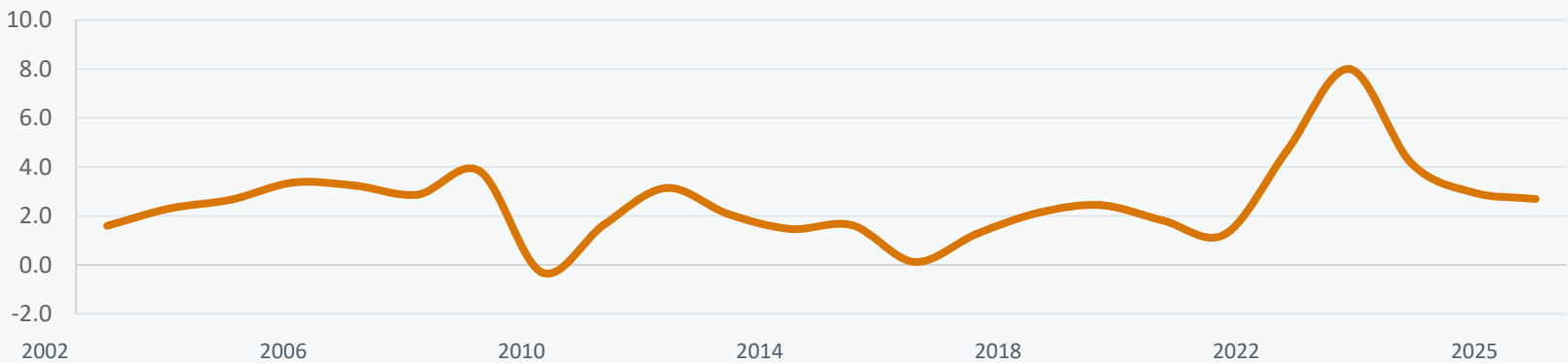
The unemployment rate indicates labor-market conditions;
maximum employment has no fixed unemployment cutoff.

The funds rate shows the policy response. Comparing dated series reveals sequence and possible delays, while leaving other causes open.

Crises interrupt periods of low inflation and unemployment

Annual averages of monthly observations, 2002–2025

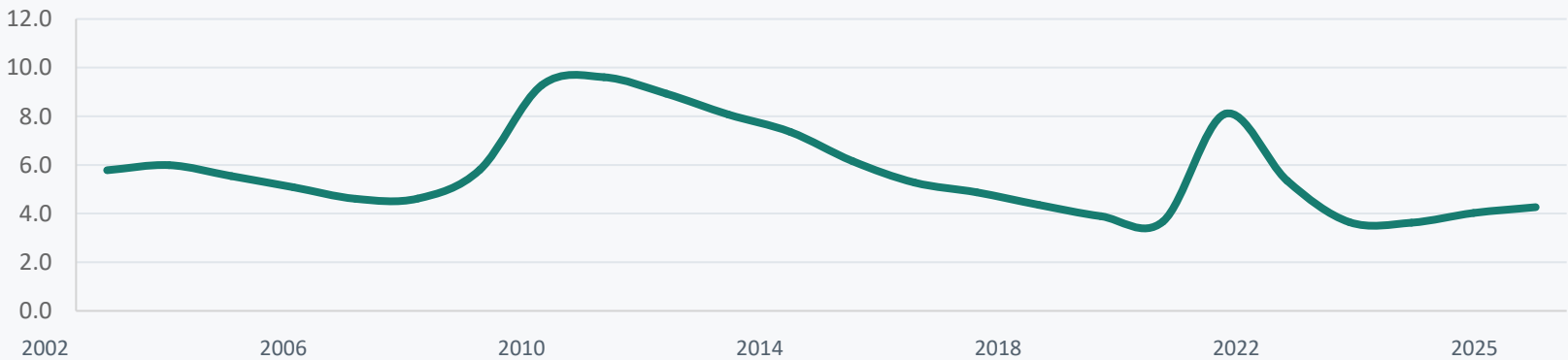
CPI inflation



2008–09

Unemployment rises after the financial crisis.

Unemployment rate

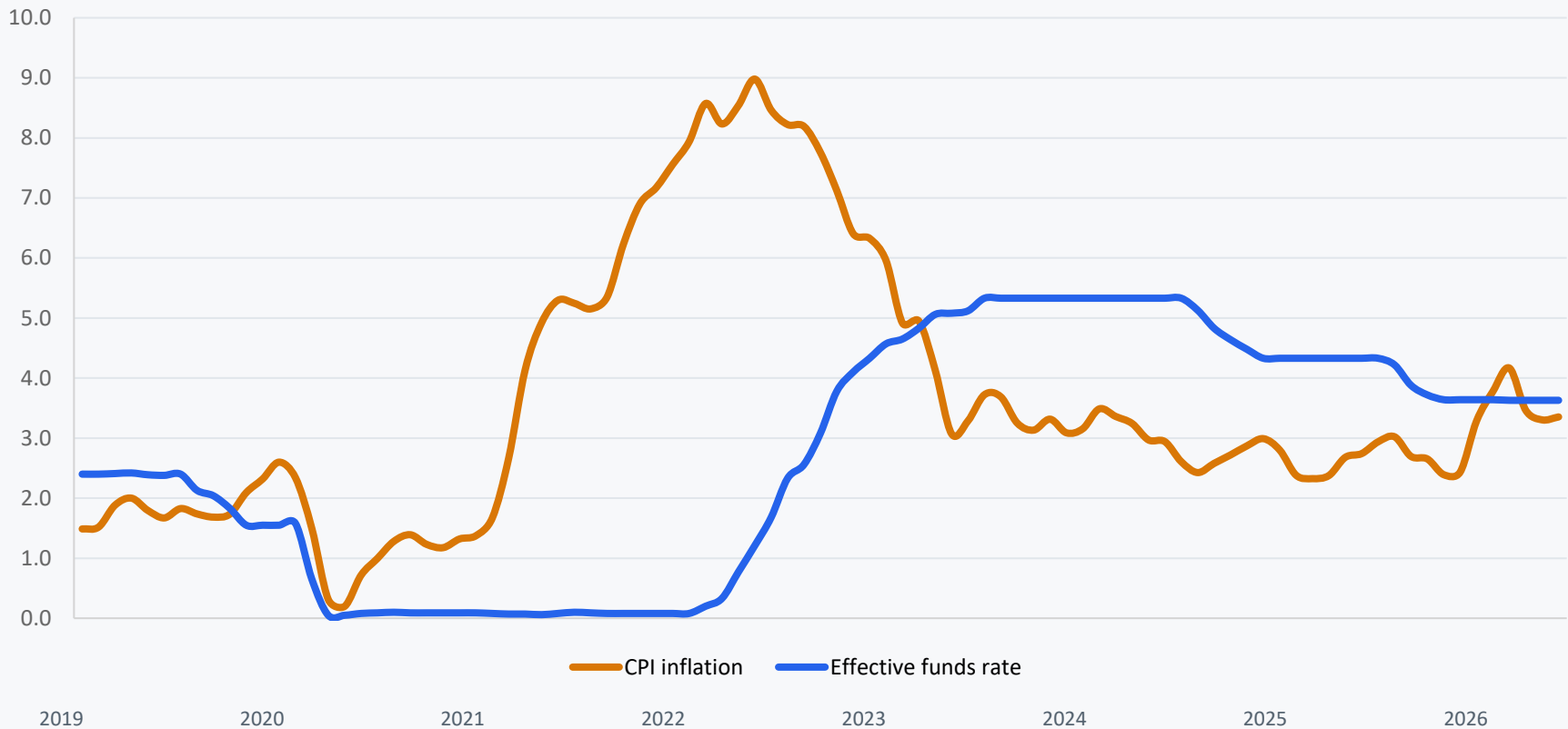


2021–22

Inflation surges after the pandemic downturn.

Inflation rose before the major rate increases

Monthly CPI inflation and effective federal funds rate, January 2019–August 2026



9.0%

CPI inflation
June 2022

Rate increases followed the inflation surge. The lines do not prove how much the hikes changed prices.

October 2025 is omitted. This slide joins adjacent observations; the notebook chart shows the gap.

Unemployment recovered after the pandemic shock

Monthly unemployment rate, January 2019–August 2026



14.8%

April 2020
pandemic peak

4.1%

August 2026

Why a separate panel: the 2020 job-loss spike needs its own scale; unemployment is not a causal test of rate hikes.

The evidence supports a qualified answer

06

1

Both goals were achieved in some periods

Low inflation and low unemployment coexisted. Neither remained stable throughout the 25-year window.

2

Recent disinflation came with modest unemployment

After the 2022 rate increases, inflation fell while unemployment stayed far below its 2020 peak. Inflation rose again in spring 2026.

What the charts cannot establish

They cannot isolate the Fed's effect from supply shocks, fiscal policy or global events.

Low unemployment does not by itself prove the economy reached maximum employment.

The 2% formal target applies to PCE inflation; the assignment specifies CPI.

Sources, visual choices and reproducibility

07

Why these visuals

The 25-year aligned charts show when the two mandate measures moved together or diverged.

The monthly inflation and rate chart reveals order and delay. A separate unemployment panel makes the pandemic shock readable.

Plotly produced the Story 2 visualizations. Story 1 used Matplotlib and Seaborn, meeting the assignment's different-library requirement.

Sources and method

BLS CPI-U: CUSR0000SA0
BLS unemployment: LNS14000000

FRED: FEDFUNDS (effective rate)

The notebook includes coded BLS API requests, retrieval messages, source fallback, and the monthly table.

[AI-use disclosure is provided in the separate accompanying file.](#)
[Supporting notebook, processed data, and reproducible analysis are available through the GitHub repository linked in the submission comments.](#)